

PART E • ALGOPROFESSOR 30-DAY INDUSTRY PROGRAM

Neurons to Mixture of Experts: Deep Learning Mastery Bootcamp

*ANN • CNN • RNN • LSTM • Transformer • MoE – with real-time industry and UN SDG capstone projects***Program Director:** Dr S Satyanarayana, CEO & Chief Agentic AI Scientist, AlgoProfessor AI R&D Solutions

30 days • 5 modules • 30 hands-on labs • 1 SDG capstone

Module 1 • Days 1–6

Foundations & ANN

Module 2 • Days 7–12

CNN & Computer Vision

Module 3 • Days 13–18

RNN, LSTM & Time Series

Module 4 • Days 19–24

Transformers, LLMs & MoE

Module 5 • Days 25–30

MLOps & SDG Capstone

Who should attend

- B.Tech / M.Tech / MCA students and final-year project teams
- Faculty members and Ph.D. scholars in AI, CSE and data science
- Software engineers moving into AI / ML roles
- Industry R&D teams building real-time AI systems
- Startup founders working on SDG-impact products

Learning outcomes

- Derive and explain the mathematics of all six architectures
- Implement each one in PyTorch from first principles
- Choose the right model for tabular, image, sequence and text data
- Fine-tune pre-trained Transformers, LLMs and MoE models
- Deploy and monitor a real-time AI system with MLOps tools
- Deliver an SDG-aligned capstone ready for publication or a pilot

What you receive

- This complete reference handbook (Parts A–E)
- Six tested PyTorch programs with sample outputs
- 60 project briefs with solution hints
- 17 SDG industry blueprints and the production-stack guide
- Personal capstone review by the program director

Prerequisites: basic Python and school-level linear algebra, calculus and probability.

PROGRAM FEE

INR 1,00,000

per participant • 30-day program

Enrol / enquire

Email: ceo@algotprofessor.com**Web:** www.algotprofessor.com**AlgoProfessor AI R&D Solutions**
Learn it. Build it. Deploy it for impact.

Table 22 • 30-Day Day-by-Day Schedule

Every day: concept session + hands-on lab

Day	Topic	Hands-on lab / outcome	Handbook reference
Module 1: Foundations & ANN (Days 1–6)			
1	Environment setup, tensors, autograd, GPU vs CPU	Install the Table 7 stack; first tensors and gradients in PyTorch	Table 7, setup listing
2	Neuron, activations, loss functions, backpropagation	Compute one neuron and one backprop step by hand, then check in code	Tables 2–3
3	Multi-layer perceptron end to end	Build and train <code>ann_mlp.py</code> ; read input → output demo	Listing B.1, Table 10
4	Regularisation, optimisers, the proper way	Dropout, weight decay, AdamW, schedulers; apply 14-point checklist	Table 9
5	ANN project lab	Credit-card fraud or diabetes risk with PR-AUC and SHAP	Table 12
6	Module 1 assessment	Quiz + mini-project demo and code review	–
Module 2: CNN & Computer Vision (Days 7–12)			
7	Convolution mathematics	Output-size, parameter and MAC calculations; receptive field	Tables 1–4, Fig. 1b
8	Building a CNN	Train <code>cnn.py</code> on MNIST; inspect feature maps	Listing B.2
9	Transfer learning	Fine-tune ResNet / EfficientNet (<code>timmm</code>) on a medical or agri dataset	Table 13
10	Detection and segmentation	YOLOv8 object detection and U-Net segmentation	Table 13
11	Explainability and edge deployment	Grad-CAM; export to ONNX / TFLite; measure FPS	Table 8
12	CNN project lab	Diabetic-retinopathy grading or leaf-disease mobile app	Table 13
Module 3: RNN, LSTM & Time Series (Days 13–18)			
13	Recurrent networks and BPTT	Vanishing-gradient experiment (0.9^{50} vs 0.99^{50})	Tables 2–3, Fig. 1c
14	RNN lab	Train <code>rnn.py</code> ; compare with naive baseline	Listing B.3
15	LSTM gates and memory	Trace one LSTM cell by hand; train <code>lstm.py</code>	Fig. 1d, Listing B.4
16	Time-series forecasting practice	Walk-forward validation, scaling without leakage, multi-step forecasts	Table 15
17	Seq2seq and attention	Encoder–decoder LSTM with Bahdanau attention	Table 15
18	Sequence project lab	Electricity-load forecasting or log anomaly detection	Table 15
Module 4: Transformers, LLMs & MoE (Days 19–24)			
19	Self-attention and the Transformer	Implement attention from equations; train <code>transformer.py</code>	Fig. 1e, Listing B.5
20	Pre-trained Transformers	Fine-tune BERT with Hugging Face <code>Trainer</code>	Listing B.7, Table 16
21	Vision, speech and multimodal Transformers	ViT for X-rays, Whisper for Indian-language speech	Table 16
22	LLMs, RAG and parameter-efficient fine-tuning	Build a RAG chatbot; LoRA / QLoRA with PEFT	Table 16
23	Mixture of Experts	Router, top- <i>k</i> dispatch, balancing loss; train <code>moe.py</code>	Fig. 1f, Listing B.6
24	Serving large MoE models	Mixtral / DeepSeek-MoE with vLLM; tokens/s and memory study	Table 17
Module 5: MLOps & SDG Capstone (Days 25–30)			
25	Real-time MLOps stack	Kafka/MQTT ingestion, MLflow tracking, FastAPI / Triton serving	Table 20, Fig. 5
26	Responsible and trustworthy AI	SHAP, Fairlearn, differential privacy, federated learning; DPDP Act 2023	Tables 18, 20
27	SDG capstone: kick-off	Choose an SDG project, define KPI, build data pipeline and baseline	Tables 18–19
28	SDG capstone: modelling	Train the hybrid CNN / LSTM / Transformer / MoE system	Table 18
29	SDG capstone: deployment	Deploy as Streamlit app or API; measure target KPI	Table 21
30	Demo day and roadmap	Capstone presentation, expert review, publication / patent / career roadmap	Part D

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